Closure of Attribute sets: X+

Attribute Closure X+ of an Attribute set X $\longrightarrow$ set of atts which can be functionally determined from it.

purpose ↦ determine if a given att. set is a candidate key or not.

↦ " all possible CKs for testing All NFs

Superkeys:

A set of Att(X) whose closure (X+) contains all the atts of the given rltn.
(prev defn: uniquely identify each record)

Candidate Keys:

A super key whose proper (at least 1 element less) subset is not a superkey. (no smaller SK)

(prev defn: minimum SK)

Computation Method 1:

Using Armstrong's Axioms without any optimization.

Trick: Compute all $\alpha \rightarrow \beta_i$ for all possible i.
→ Union them to get α^+

Given R(A, B, C, D, E)

FD: $A \rightarrow B$
$B \rightarrow C$
$C \rightarrow D$
$D \rightarrow E$

∴ $A \rightarrow C$ (transitivity)
$A \rightarrow A$ (reflexivity)
$A \rightarrow D$ (transitivity)
$A \rightarrow E$ (")

$A \longrightarrow A, B, C, D, E$ (union)

So, $A^+ = \{A, B, C, D, E\}$

Now, $B \rightarrow D$ (trans)

$B \rightarrow E$ (")

$B \rightarrow B$

$B \rightarrow B, C, D, E$ (U)

So, $B^+ = \{B, C, D, E\}$

Again, $C \rightarrow E$ (trans)

$C \rightarrow C$

$C \longrightarrow C, D, E$

So, $C^+ = \{C, D, E\}$

Method 2 : simplified · if $\alpha \rightarrow \beta$ holds & α already in set, add β.

$A^+ = \{A, B, C, D, E\}$ SK ✓ CK ✓

$AD^+ = \{A, D, B, E, C\}$ SK ✓ (A is proper subset of AD so not CK) ✓

$AB^+ = \{A, B, C, D, E\}$ SK

$B^+ = \{B, C, D, E\}$ ✗ (A नाही)

$CD^+ = \{C, D, E\}$ ✗ (AB ")

$C^+ = \{C, D, E\}$ ✗ (")

So, A is the only Candidate key.

Method 3 :

Given $R(A, B, C, D, E)$

FD: $A \rightarrow B$

$D \rightarrow E$

$A^+ = \{A\}$
$= \{A, B\}$

$BC^+ = \{B, C\}$?

$ABCDE^+ = \{$ all $\}$ ✓

$ABDE^+ = \{A, B, D, E\}$ ✗

$ACDE^+ = \{A, C, D, E, B\}$ ✓

$ACD^+ = \{A, C, D, B, E\}$ ✓ SK.



Method 3 (example) Given: $R(A, B, C, D, E)$

(step 1 ← discard step) FD $\{A \to B,\ D \to E\}$

Lets start with A, B, C, D, E (all)

$ABCDE^+ = \{A, B, C, D, E\}$

Now, FD, $A \to B$ so, B can be removed from Left side and still will be able to reduce all attributes. (A is already there.)

$ABCDE^+ = \{A, C, D, E, B\}$ (RR) $A \to B$

$\Rightarrow ACDE^+ = \{A, B, C, D, E\}$

apply $D \to E$ (both are there so one can be removed — Right part)

$\Rightarrow ACDE^+ = \{A, C, D, B, E\}$

$ACD^+ = \{A, B, C, D, E\}$ (SK) ✓

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Closure of Attributes | Normalization | Fist Normal Form and Atomic Domain | Second Normal Form | Third Normal Form | Boyce - Codd Normal Form (BCNF)

Method 3: Simple Example Cont.

So, $ACD^+ = \{A, B, C, D, E\}$ (SK) ✓

Need to test all proper sub sets are SK or not to finally say ACD is a C.K.

$\{A \to B,\ D \to E\}$ FD

ACD all proper subsets are:

$2^3 - 2 = 6$ (No.)

AC	need to check
CD	"
AD	"
A	"
C	"
D	"

$AC^+ = \{A, C, B\}$ × SK

$CD^+ = \{C, D, E\}$ × SK

$AD^+ = \{A, D, E, B\}$ × SK

$A^+ = \{A, B\}$ × SK

$C^+ = \{C\}$ × SK

$D^+ = \{D, E\}$ × SK

Note: Here we have (6) proper subsets to check in contrast of earlier version $ABCDE = 2^5 - 2 = 30$

"step1 motivation is to reduce the no. of checks."

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Method 3: Simple Example Cont.

Method 3 : step 2

A New Term - Prime Attributes (PAs) : Collection of attributes of all candidate Keys of a given relation.

In step 2 we got ACD as CK (by testing ACD^+)

So, upto now PAs = {A, C, D}

Principle : For a given CK, form another set called Dependent set (DS) [e.g. $A \to B$, $D \to E$ ⇒ B, E] also called Dep Attributes (DA)

IF $PA \cap DA = \phi$ then CK is the only CK (present example) no other possible CK is there.

Else go for the Replacement Process (RP) (will later be explained) (Now example will be given)

END IF;

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Method 3: Simple Example (End)

Given $R(A, B, C, D, E)$ FD $(A \to B,\ D \to E)$

So, $DA = \{B, E\}$ and $PA = \{A, C, D\}$ (so upto now CK = ACD)

So, $PA \cap DA = \{A, C, D\} \cap \{B, E\}$

$= \phi$

So, ACD is the only CK (Proved)

(Fixed) ACD was CK in step 1

$R = (A, B, C, D)$ with $A \to B$, $B \to C$, $C \to A$

FD: $A \to B$, $B \to C$, $C \to A$

$AD^+ \longrightarrow SK$

Check all proper subset.

$A^+ \to ABC$ ✗

$D^+ \to D$ ✗

$AD \to CK$.

$PA \to (A, D)$

(fixed) $DA \to (A, B, C)$

$PA \cap DA = A$

$C \to AD$

$\downarrow$

CD $\to (CK)$

$PA \cap DA = C$

$B \to CD$

$\downarrow$

BD

$\downarrow$

AD (already checked)

∴ set of cks (AD, BD, CD)

2 R(A, B, C, D)

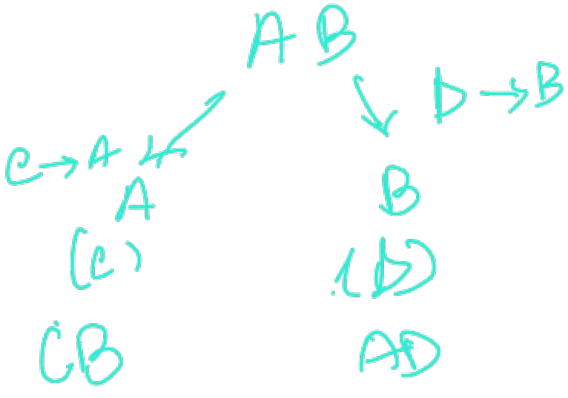
AB → CD D → B , C → A

AB+ → CK

PA = {A, B}.

DA = {C, D, B, A}

∩ = {A, B}



AB → CD, D → B ∴ AD → CD

last check:

AB CD → All

CD+ → All

∴ CK = {AB, BC, CD, AD}

2NF: R(A,B,C,D E,F)

FD: A→B, B→C, C→D, D→E

AF⁺ → CK.

PA = {A, F}

DA = {B, C, D, E}

AF is the only CK.

So PSS(CK) → NON Prime Ar.

Not 2NF.

Hk R(A,B,C,D), FD: AB→CD, C→A, D→B

{AB, AD, BC, CD}

A, B, D, C ✗

So in 2NF

3NF too!
NonPA.

H.NS R(A,B,C,E,F)

AB → C,D,E,F

BD → F

↓
NonPA.

$AB^+ \rightarrow CK.$

$PA = (A, B)$

$\cap = \emptyset$ AB only CK.

NOT 3NF!

D NonPA
so BD as a
whole NonPA

BCNF: Multiple overlapping CKs.

$R(A, B, C)$ $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow A$

(A, B, C)